Track A : Cardiovascular Decision Support System: Documentation

1. Overview and clinical-safety statement

The system turns 50 routine clinical inputs into four probability estimates (overall CAD and LAD, LCX and RCA stenosis of 50% or more) and shows them on an interactive 3D heart with a per-patient explanation of the factors behind each estimate.

It is built on 303 patient records and a frozen model set (v1.0.0): **logistic regression** for CAD, LAD and RCA, and **XGBoost** for LCX. A FastAPI backend serves the predictions and explanations; a React Three Fiber frontend colours each artery on a fixed 0 to 100% scale.

Clinical-safety disclaimer. This is a research prototype for a hackathon. It is not a medical device, gives no diagnosis and must not guide treatment. All outputs are statistical estimates from a small single-centre referral cohort and need review by a qualified clinician.

- Live demo: https://track-a-frontend-phi.vercel.app/ ***(the free backend may take 30 to 60 seconds to wake on the first request)***
- Backend repository: tracka-backend)
- Frontend repository: tracka-frontend)
- Demo video: YouTube link added

2. Data and preprocessing

All models are trained on one public dataset of 303 patients (the extended Z-Alizadeh Sani cardiac catheterisation cohort), with no external data. Class counts are 216 CAD-positive, 177 LAD, 119 LCX and 114 RCA; a vessel is positive at 50% or more narrowing.

The API takes 50 inputs in six groups; BMI is computed on the server from weight and height.

Group	Inputs
Demographics	4
History and risk factors	11
Examination and symptoms	11
ECG	7
Laboratory	14

Echocardiography 3

Preprocessing is deliberately minimal and identical in training and serving:

- Every field is required; nothing is imputed, in training or in serving.
- Logistic regression models carry their own StandardScaler; XGBoost uses unscaled values.
- Chest-pain flags and bundle-branch block are exposed as one enum each and expanded to the dataset's binary columns.
- Leakage guard: LAD, LCX, RCA and Cath are never inputs. Exertional CP (constant) and Obesity (a BMI threshold) were dropped.
- The source typo "Fmale" is read as female.

The model card records the dataset SHA-256, library versions and random seeds, so a run can be reproduced from the repository.

3. Models and validation methodology

Each target has its own independent model: L2 logistic regression for CAD, LAD and RCA, and XGBoost for LCX. Logistic coefficients give exact, auditable explanations; XGBoost is explained with TreeSHAP.

Validation follows one protocol throughout:

- 5x5 repeated stratified cross-validation (five repeats of five folds).
- Hyper-parameters are tuned inside each training fold (nested), so test folds never influence tuning.
- Confidence intervals are patient-level bootstrap intervals on out-of-fold predictions. The model card flags these as optimistic.
- Probabilities are used raw: post-hoc calibration did not help in nested CV.
- Final hyper-parameters are fitted on all 303 rows; the reported figures come from nested CV, not from that final fit.

Four improvement ideas for LCX and RCA (six variants) were tested against pre-declared acceptance gates and all rejected, so the baseline stayed frozen: CAD-conditional models, interaction terms, a shared PLS latent factor, and multi-output XGBoost. The largest change in AUC was -0.039, and no variant gained more than +0.005.

4. Results

CAD discrimination is high (AUC 0.925), LAD is moderate (0.830), and LCX and RCA are limited (0.734 and 0.712). All four beat the no-skill Brier score.

[embed: node/c8229a2b-339e]

Target	Model	Prevalence	Brier (no-skill)	Calibration slope	ECE (5 bins)	AUC within CAD-positive
CAD	LR	71.3%	0.098 (0.205)	1.13	0.022	not applicable
LAD	LR	58.4%	0.163 (0.243)	1.15	0.044	0.618
LCX	XGBoost	39.3%	0.204 (0.239)	0.92	0.041	0.624
RCA	LR	37.6%	0.207 (0.235)	0.87	0.033	0.576

At reference thresholds that the API does not use, CAD reaches precision 0.904, recall 0.908 and F1 0.906 (threshold 0.54, max-F1 rule). LAD reaches precision 0.773, recall 0.879 and F1 0.822 (threshold 0.48, Youden rule). Both come from nested CV. No thresholds are set for LCX and RCA, and the API returns probabilities only, never a hard label.

Vessel-level signal is weak: among CAD-positive patients the AUC is only 0.58 to 0.62, so LAD, LCX and RCA estimates partly track overall CAD risk. The interface says so next to each vessel. Accuracy was not recorded in the frozen validation, so this report does not quote it; AUC, Brier score and calibration are the headline measures because the cohort is imbalanced.

5. 3D visualization

The viewer shows an anatomical heart whose left anterior descending (LAD), left circumflex (LCX) and right coronary (RCA) arteries are each coloured by that patient's predicted stenosis probability, so risk is read directly from where it sits on the heart.

- **Anatomy.** The heart is built from named BodyParts3D parts (FMA ids), grouped into the three arteries, the left main stem, ventricles, atria and great vessels. No artery position was guessed from geometry. A procedural schematic heart is available as a toggle and as an automatic fallback if the model fails to load.
- **Spatial risk mapping.** Each artery takes one uniform colour from a fixed viridis scale of 0 to 100%, so colours mean the same thing for every patient. Probabilities are never clamped or rescaled. Before an assessment, vessels are neutral grey. The left main stem and heart wall stay uncoloured because the models do not predict them. CAD has no vessel, so overall risk is shown as a score beside the heart.

- **Interaction.** Orbit, zoom and four view presets (front, patient's left, patient's right, back). Clicking an artery opens its probability, discrimination tier and explanation. Labels follow the camera and hide when their artery faces away, so a label never points at the wrong side.
- **Honest framing.** A legend states the fixed scale, and a credit line cites the source. The notice that arteries are not patient-specific anatomy is always visible.

Light and dark themes are both supported.

6. Clinical interpretability

Every estimate comes with the ten input factors that moved it most, computed exactly from the frozen model rather than approximated by a surrogate.

- **Logistic models (CAD, LAD, RCA).** Each factor's contribution is its coefficient times the patient's standardised deviation from the training mean. Contributions add up to the model's raw score; the reference is the intercept, the score for an average training patient.
- **XGBoost (LCX).** Contributions are exact TreeSHAP values (the model's own pred_contribs), which also add up to the raw score.
- **Display.** A bar chart on the explanation page (#/explain) shows factors that raise the estimate in orange and factors that lower it in blue, with the patient's value beside each. The heading reads "Factors influencing this model estimate".
- **Wording.** Factors are described as influences on a model estimate, never as causes. Contributions are in log-odds, so they are exact for the score but not directly additive in percentage points.
- **Reliability cues.** Each vessel shows its discrimination tier (high, moderate or limited) and a caution chip for LCX and RCA. If a vessel estimate exceeds overall CAD by more than the display margin, the interface flags it and leaves the values unadjusted.

The explanation service is a separate endpoint that does not change /assess: predictions are identical with or without it, and a test checks that the contributions reconstruct the predicted probability.

7. System architecture, running, deployment and testing

[embed: node/7936f3d7-1714]

The inference core is framework-free and byte-identical to the approved file; tests check its hash. On startup the API verifies every model file against manifest.json and refuses

to start if one differs, so what is served is exactly what was validated. Every input is required and rejected with a field-level HTTP 422 if invalid.

Run locally.

1. Backend: pip install -r requirements-dev.txt, then uvicorn app.main:app --port 8000. Set TRACKA_ARTIFACT_DIR to use another artifact folder.
2. Frontend: npm install, then npm run dev. Vite proxies /api to the backend; set BACKEND_URL to point elsewhere.
3. Open the printed local address (port 5173), enter a patient and run the assessment.

Deployment. The frontend runs on Vercel and the backend on Render (free plan, Python 3.12). Both build from their own repository. The first request after idle can take 30 to 60 seconds while the backend wakes.

Testing. The backend suite has 576 tests: contract and schema checks, golden patients, parity with the reference implementation, leakage guards, input validation and explanation additivity on all 303 patients. The frontend has 124 unit and component tests, plus a Playwright end-to-end smoke test that loads the 3D scene, assesses a patient and checks colours and labels against the API values.

8. Limitations and responsible use

The estimates are only as good as a 303-patient, single-centre referral cohort allows, and they must not be used to make or exclude a diagnosis.

- **Population.** CAD prevalence is 71% in the training data, so probabilities are conditional on a high-risk referral population and will overstate risk in general populations. There is no external validation.
- **Small samples.** There are 119 LCX and 114 RCA positives, so intervals are wide. Several binary flags have fewer than 10 positives (CHF has one), so their effects are unreliable.
- **Weak vessel signal.** Among CAD-positive patients, vessel AUC is 0.58 to 0.62. LCX and RCA estimates mostly track overall CAD risk, so they are shown as coarse estimates with no hard label.
- **Optimistic intervals.** Confidence intervals are bootstrap intervals on out-of-fold predictions and the model card marks them optimistic.
- **Coarse labels.** Vessel labels mean 50% or more narrowing of the whole vessel, with no lesion location, so the 3D colours cannot show where along an artery a stenosis sits.

- **Unverified inputs.** Units for most laboratory values are inferred from value ranges, BP is a single presumed-systolic value, and Region RWMA and Function Class are integer codes whose meaning is undocumented. The models use them as plain numbers.
- **Independent models.** The four models are separate and nothing forces them to agree, so a vessel can score above overall CAD. The interface flags this and leaves the numbers as they are.
- **Illustrative anatomy.** The heart is a generic anatomical model, not the patient's own arteries.
- **Input range.** Range checks on absurd values are limited: an extreme input such as an LDL of 1,000,000 can return a server error instead of a clear validation message. Range validation is future work.

For responsible use: the system supports a clinician's reasoning and never replaces it, and the clinical-safety disclaimer stays visible in the interface.

9. References and credits

- **Data.** Extended Z-Alizadeh Sani coronary artery disease dataset, 303 patients (public; source file hash recorded in the model card). Original work: Alizadehsani et al., Computer Methods and Programs in Biomedicine, 2013.
- **3D anatomy.** BodyParts3D, Database Center for Life Science (DBCLS), licensed CC BY-SA 2.1 Japan; heart concept FMA67135. Mitsuhashi et al., Nucleic Acids Research, 2009. The derived heart model in this project is shared under the same licence; its credit file ships with the model.
- **Explanations.** Lundberg and Lee, "A unified approach to interpreting model predictions", 2017; Lundberg et al., "From local explanations to global understanding with explainable AI for trees", Nature Machine Intelligence, 2020.
- **Software.** FastAPI, scikit-learn, XGBoost, NumPy, React, Vite, three.js, React Three Fiber, Zustand, Vitest and Playwright.